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$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} + \sqrt{9x^2 - 1} - 5x)$$

opsplitsen:

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} + \sqrt{9x^2 - 1} - 5x) = \lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} - 2x) + \lim_{x \rightarrow \infty} (\sqrt{9x^2 - 1} - 3x)$$

Eerste term:

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\sqrt{4x^2 + x} - 2x \cdot \left(\frac{\sqrt{4x^2 + x} + 2x}{\sqrt{4x^2 + x} + 2x} \right) \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{(\sqrt{4x^2 + x} - 2x) \cdot (\sqrt{4x^2 + x} + 2x)}{(\sqrt{4x^2 + x} + 2x)} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{4x^2 + x - 4x^2}{\sqrt{4x^2 + x} + 2x} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{x}{\sqrt{4x^2 + x} + 2x} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{1}{\sqrt{4 + \frac{1}{x}} + 2} \right) \\ &= \frac{1}{4} \end{aligned}$$

Tweede term:

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\sqrt{9x^2 - 1} - 3x \cdot \left(\frac{\sqrt{9x^2 - 1} + 3x}{\sqrt{9x^2 - 1} + 3x} \right) \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{(\sqrt{9x^2 - 1} - 3x) \cdot (\sqrt{9x^2 - 1} + 3x)}{(\sqrt{9x^2 - 1} + 3x)} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{-1}{\sqrt{9x^2 - 1} + 3x} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{-1}{x\sqrt{9 - \frac{1}{x^2}} + 3} \right) \\ &= \lim_{x \rightarrow \infty} \left(\frac{-1}{x \cdot \sqrt{9} + 3} \right) \\ &= 0 \end{aligned}$$

$$\lim_{x \rightarrow \infty} (\sqrt{4x^2 + x} + \sqrt{9x^2 - 1 - 5x}) = \frac{1}{4} + 0 = \frac{1}{4}$$

References